The MAGI Champion Architecture

Cortico-Cerebellar Competitive Dual-Kernel Network with Locus Coeruleus Volatility and Systemic Decoupling

MAGI Automated Search Engine

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Abstract

This document formally specifies the mathematical architecture of the MAGI Champion model, derived through a massive 128-subject global evolutionary optimization (CMA-ES). The model achieved strict dominance over the abstract polynomial baseline on the generative likelihood function ($p < 0.01$). It integrates competitive Cortico-Cerebellar learning, High-Frequency Variance (HFV) via Instantaneous Shock, Pearce-Hall Locus Coeruleus Volatility, and biologically grounded Polynomial Systemic Decoupling.

1 Cortical Learning: The Primary Reinforcement Engine

The cortex maintains the primary value representations for each action $a \in \{0, 1\}$. It updates using a standard Rescorla-Wagner rule. The reward prediction error (RPE) on trial t for chosen action a is:

$$RPE_{ctx,t} = R_t - Q_{ctx,t}(a) \tag{1}$$

The value state is updated via the cortical learning rate α_{ctx}:

$$Q_{ctx,t+1}(a) = Q_{ctx,t}(a) + \alpha_{ctx} \cdot RPE_{ctx,t} \tag{2}$$

2 Cerebellar Learning: The Temporal Basis Engine

The cerebellum maps temporal context via a population of Granule Cells with log-uniform decay timescales τ_k, where $k \in \{1, \ldots, N_{MF}\}$.

2.1 State Representation

During the Inter-Trial Interval (ITI), the active basis decays:

$$x_{k,t} = x_{k,t-1} \cdot \exp\left(-\frac{ITI}{\tau_k}\right) \tag{3}$$

2.2 Purkinje Cell Integration

The Purkinje cells ($w_{k,t}$) read out this basis. Their synaptic efficacy naturally degrades over the feedback duration (F_{dur}) via leak rate λ_{gc}, while simultaneously updating based on the cerebellar error signal $RPE_{cb,t}$ and dynamic plasticity $\eta_{cb,t}$:

$$w_{k,t+1} = w_{k,t} \cdot \exp(-\lambda_{gc} \cdot F_{dur}) + \eta_{cb,t} \cdot RPE_{cb,t} \cdot x_{k,t} \tag{4}$$

The cerebellar prediction is the dot product of the basis and the weights:

$$Q_{cb,t}(a) = \sum_{k=1}^{N_{MF}} w_{k,t}(a) \cdot x_{k,t} \tag{5}$$

3 The Competitive Routing Topology

To prevent parameter unidentifiability between the slow-learning Cortex and fast-learning Cerebellum, the networks exhibit a *Competitive Dual-Kernel* topology. The Cerebellum's error signal is strictly suppressed when the Cortex is highly accurate (low $|RPE_{ctx}|$):

$$RPE_{cb,t} = (R_t - Q_{cb,t}) \cdot (1 - |RPE_{ctx,t}|) \tag{6}$$

4 Locus Coeruleus (LC) Volatility & Arousal

The Locus Coeruleus acts as an uncertainty accumulator, filtering the high-frequency cerebellar absolute error via a Pearce-Hall mechanism to generate a volatility scalar Ω_t:

$$\Omega_t = (1 - \alpha_{vol})\Omega_{t-1} + \alpha_{vol}|RPE_{cb,t}| \tag{7}$$

This scalar dynamically dilates cerebellar plasticity (η_{cb}), allowing the cerebellum to rapidly acquire new temporal bases during environmental shocks:

$$\eta_{cb,t} = \eta_{base} + \beta_{ph} \cdot \Omega_t \tag{8}$$

5 The Drift-Diffusion Model (DDM) Interface

The integrated network utility drives the parameters of the behavioral Drift-Diffusion Model.

$$\Delta U_t = (Q_{ctx}(1) + Q_{cb}(1)) - (Q_{ctx}(2) + Q_{cb}(2)) \tag{9}$$

5.1 Drift Rate (v_t): Arousal Suppression

The drift rate is proportional to the difference in utility, but is exponentially suppressed by systemic volatility, representing hesitation during highly uncertain environmental blocks:

$$v_t = \kappa_v \cdot \Delta U_t \cdot \exp(-\beta_{vol\_v} \cdot \Omega_t) \tag{10}$$

5.2 Boundary (a_t): Shock & Systemic Decoupling

The decision boundary a_t incorporates three massive structural components to maximize both probability density and first-passage time variance:

1. **High-Frequency Variance (Instantaneous Shock):** The boundary reacts instantaneously to absolute cortical surprise from the prior trial $|RPE_{ctx,t-1}|$.

2. **Low-Frequency Volatility:** The boundary dilates according to the integrated LC volatility Ω_t.

3. **Systemic Decoupling (Bio-Energetic Depletion):** The boundary drifts globally across normalized trial time $\hat{t} = t/T$ via polynomial fatigue parameters (representing receptor desensitization and adenosine accumulation).

$$a_t = a_0 + \beta_{shock}|RPE_{ctx,t-1}| + \beta_{vol\_a}\Omega_t + \beta_{p1}\hat{t} + \beta_{p2}\hat{t}^2 + \beta_{p3}\hat{t}^3 \tag{11}$$

Boundary components are thresholded to $a_t \in [0.01, 10.0]$ to guarantee biological convergence.

5.3 First-Passage Time Distribution

The response choices ($y \in \{1, 2\}$) and reaction times (RT) are evaluated over the continuous Wiener first-passage time probability density function (WFPT):

$$RT \sim \text{Wiener}(v_t, a_t, w = 0.5, t_{nd}) \tag{12}$$